FRONTIER PROTOCOL

Game Design Document • v0.1

A sci-fi colony roguelite with isometric exploration, persistent scouts,

a living alien planet, and a factory-driven meta game.

Genre	Target Platform	Target Price
Roguelite / Strategy	PC (Steam)	€9.99 – €14.99

1. Concept Overview

Frontier Protocol is a sci-fi roguelite in which players direct scouting parties across a procedurally generated alien planet, returning home between expeditions to grow a colony, operate a resource factory, and prepare for the next push into hostile wilderness.

The game is built around two interlocking compulsion loops:

- The Run — a tense isometric expedition into unknown territory with a squad of up to four named scouts.
- The Meta — a colony-building, factory-management layer where every resource gathered and every discovery made translates into permanent progress.

> **VISION** *The core fantasy is going from desperate survivor to confident colonizer over the course of a full campaign — and doing it again, differently, in New Game Plus.*

1.1 Elevator Pitch

XCOM's soldier permadeath and emotional investment. Hades' run-to-meta momentum loop. Civilization's 'one more turn' compulsion. The alien wonder of a newly discovered planet. Wrapped in an isometric roguelite that you can finish in 15–20 hours and immediately want to replay.

1.2 Key Pillars

- Every run matters — scouts level up, the map changes, the colony advances. A failed run is never wasted.
- Characters, not units — your scouts have names, traits, relationships, fears and histories. Their deaths mean something.
- A living world — the planet is not static. Fauna migrates, threats escalate, and the wilderness pushes back.
- Ethical depth — systems reward understanding and mastery, never spending. No loot boxes, no pay-to-win.
- Achievable scope — designed from the ground up for a focused solo developer with the intent to ship.

2. Core Loops

2.1 The Three Loops

Frontier Protocol operates on three nested time horizons, each feeding the next:

Loop	Duration	Core Activity	Reward
Short — The Run	20–40 min	Scout expedition, combat, discovery	Resources, scout XP, map data
Medium — The Session	1–2 hrs	Colony building, factory management, squad prep	Unlocks, stronger expeditions, narrative beats
Long — The Campaign	15–20 hrs	Reveal the planet, overcome its central threat	Story resolution, NG+ modifiers, true endings

2.2 The Short Loop — Expedition

A run follows a clear rhythm: equip your squad at the colony → push into the procedural map chunk by chunk → gather resources, encounter events, fight creatures → decide when to retreat before your supplies or morale collapse → debrief at the colony.

The tension dial is supply-based. Scouts carry oxygen, rations, and medical supplies. Deeper exploration finds better rewards but burns through supplies faster. The decision to turn back is always uncomfortable — there is always something just over the next ridge.

2.3 The Medium Loop — Colony Session

Between expeditions the player is in the colony interface. Multiple things are always in motion simultaneously — the factory is processing the last haul, a building is under construction, a scout is recovering from an injury, a new recruit has arrived with an unusual trait profile. There is never a clean stopping point. Something is always almost ready.

> **DESIGN NOTE** *This is the Civilization 'one more turn' mechanism. The player should always have at least three things they want to see resolve before they quit.*

2.4 The Long Loop — Campaign Arc

Each campaign has a macro goal — establishing the colony to a point where the main colony ship can safely land. This requires pushing into increasingly dangerous zones of the planet, confronting a central escalating threat, and building the colony infrastructure to support endgame expeditions.

New Game Plus campaigns offer modifiers that change faction behaviours, starting conditions, and planet generation parameters — preserving replayability after the first story resolution.

3. The Expedition

3.1 Map Structure — Chunked Exploration

The planet surface is divided into procedurally generated chunks, each roughly one to two screens in size. Chunks are assembled from typed templates — jungle, rocky mesa, wetland, crystalline flats, alien ruin, crater — and populated with content according to their type and current world state.

Within a chunk, movement is free. Scouts walk to points of interest, interact with flora and fauna, harvest resource nodes, and trigger encounters by proximity. There is no action point grid. The feel is closer to Divinity: Original Sin's exploration than to XCOM's tactical maps.

- Adjacent chunks are visible before entry, giving the player a rough sense of what lies ahead.
- Chunk geography is seeded and stable — the cliff face is always there. Chunk activity is rolled fresh each expedition.
- Cleared chunks gain a 'surveyed' state — scouts move faster, resource locations are known, risk is lower.

> **DESIGN NOTE** *Familiarity should feel like earned expertise, not repetitive grind. A revisited chunk is a known quantity the player can navigate confidently — but what's in it today is always a question.*

3.2 Chunk Variation System

Each chunk combines four independent variation layers to ensure revisiting never feels identical:

Layer	Stability	Examples
Geography	Permanent (seeded)	Terrain shape, landmarks, resource deposit locations
Activity	Rerolled each expedition	Creature migration, rival scouts, spore bloom, storm aftermath
Event	Driven by meta state	Cleared nest now occupied, artifact site attracting rivals
Temporal	Time of day / season cycle	Night visibility, wet season flooding, meteor debris fields

3.3 Encounter Types

- Combat — creature or rival faction engagement. Positional but not grid-strict. Cover, flanking, and elevation matter.
- Discovery — artifact, crashed probe, alien structure. Triggers an event card with decisions and consequences.
- Environmental — hazard zone requiring navigation choices. Toxic pocket, unstable ground, bioluminescent trap.
- Resource Node — harvest point with optional risk. Richer nodes are often guarded or time-pressured.
- Scout Event — character-specific narrative moment triggered by a scout's trait or history.

- Faction Encounter — non-hostile contact opportunity with a rival group. Trade, information, or conflict.

3.4 Supply and Retreat

Each expedition begins with a finite supply loadout configured at the colony — oxygen canisters, rations, medkits, and mission-specific equipment. Supply consumption is steady with spikes on combat and environmental hazards.

When to retreat is always a genuine decision. Supplies can be partially replenished from resource nodes in the field. Scouts can push on low supplies but risk arriving back in critically poor condition or, in extremis, not arriving back at all.

4. The Scout System

4.1 Philosophy

Scouts are the emotional core of Frontier Protocol. They are not units — they are characters. Each one is procedurally generated but feels hand-crafted through the accumulation of experience, trauma, relationships, and story. The player should feel the loss of a veteran scout the way an XCOM player feels the loss of a soldier named after their best friend.

4.2 Scout Generation

New scouts arrive at the colony as recruits with a procedurally generated profile:

Attribute	Nature	Detail
Name & background	Flavour + mechanical seed	Background affects starting trait pool
Base stats	4–5 core values	Endurance, Precision, Awareness, Resolve, Ingenuity
Starting traits	1–3 at generation	Permanent personality and situational modifiers
Specialisation seed	Hidden	Determines which advanced traits become available on level-up

4.3 Scout Progression — The Lifecycle

Each scout progresses through a full lifecycle across the campaign:

Recruit

Fresh arrival. Unknown potential. Cheap to field, fragile, but has a trait profile that hints at where they might excel. Veteran scouts comment on promising recruits.

Field Scout

Gains experience through successful expeditions. Levels up after milestone run counts. Each level-up offers a choice from a small pool of traits — never a stat menu. The traits available are influenced by what the scout has actually experienced: a scout who has survived multiple creature attacks is offered combat resilience traits; one who has spent time in jungle biomes develops flora-reading abilities.

Veteran

A scout who has survived many expeditions accumulates scars — mechanical quirks from past trauma that are simultaneously drawbacks and marks of character. A veteran who barely survived a creature ambush might have a phobia of that creature type that imposes a Resolve penalty near them, but also a combat bonus from hard-earned familiarity. These scars make veterans feel genuinely individual.

Retirement

Scouts who accrue too many critical injuries, reach a campaign milestone, or whose player chooses to stand them down transition to colony roles. A retired scout does not disappear — they become an NPC with mechanical weight:

- Former combat specialists become drill instructors — new recruits gain a starting XP bonus.
- Former medics staff the medical bay, improving recovery times for the whole roster.
- Former cartographers work in the survey office, improving map data quality from expeditions.
- Highly decorated veterans may unlock unique colony projects or events.

Legacy

When a scout dies, their name is added to a memorial in the colony. Their strongest trait may pass to a younger scout they mentored. Buildings can be named in their honour. The colony remembers.

4.4 Party Dynamics

Expeditions are undertaken by parties of up to four scouts. Party composition creates emergent dynamics:

- Bond — scouts who run together repeatedly develop a bond granting passive bonuses when paired. Losing a bonded partner triggers a Resolve crisis.
- Rivalry — sometimes two scouts generate friction, creating small friction penalties but also competitive drive bonuses.
- Mentorship — a veteran and a recruit in the same party accelerates the recruit's trait development.
- Morale — parties track a shared morale stat during the run. It falls on casualties, scary events, and prolonged danger. It rises on discoveries, victories, and rest events.

4.5 Injury, Capture, and Rescue

Scouts are not simply alive or dead. Outcomes exist on a spectrum:

State	Duration	Consequence
Light Wound	1–2 expeditions	Reduced stats. Recovers with rest.
Serious Injury	3–5 expeditions	Requires medical bay treatment. May gain a scar trait.
Missing in Action	Timed window	Rescue mission available. Scout is recoverable.
Captured	Timed window	Rescue mission required. Difficulty scales with captor faction.
Dead	Permanent	Added to memorial. Partial legacy effects apply.

> **DESIGN NOTE** *Rescue missions are a special expedition type with a known destination, a ticking timer, and enormous emotional stakes. The player is not rolling randomly through a dungeon — they are going to get their person back.*

5. The Colony Meta

5.1 Overview

The colony is the persistent world the player returns to between expeditions. It is the Civilization layer of the game — always in motion, always presenting decisions, always showing visible progress toward multiple goals simultaneously. It should never feel like a loading screen between runs.

5.2 The Factory System

Resources gathered on expeditions feed a colony factory that processes raw materials into advanced components, equipment, medicine, and construction materials. The factory runs continuously — even during an expedition, the factory is working.

The factory operates on a simplified production chain model inspired by games like Satisfactory and Arknights: Endfield, but significantly reduced in complexity. The goal is interesting decisions about what to prioritise, not complex logistics puzzles.

- Raw resources (minerals, biological samples, alien compounds) are harvested in the field.
- Processing buildings convert raw resources into intermediate goods.
- Advanced buildings convert intermediate goods into equipment and upgrades.
- Players configure production priorities before an expedition and return to collect results.

> **DESIGN NOTE** *The factory should give the player something to come home to. Coming back from a good run to find the factory has been busy — equipment ready, resources stockpiled — should feel like arriving at a productive home base.*

5.3 Colony Buildings

Building	Category	Function
Landing Pad	Core	Starting structure. Scout deployment and return.
Medical Bay	Support	Scout recovery. Upgrades reduce recovery time and improve injury outcomes.
Survey Office	Intelligence	Processes map data. Improves chunk preview information.
Research Lab	Progression	Analyses discoveries. Unlocks new equipment and building types.
Processing Plant	Factory	Converts raw resources into intermediate goods.
Fabrication Bay	Factory	Converts intermediate goods into equipment and components.
Barracks	Scout	Increases active roster size. Enables mentorship pairing.
Rec Room	Scout	Improves morale recovery between expeditions.
Memorial Hall	Legacy	Displays fallen scouts. Unlocks legacy events.

Building	Category	Function
Signal Tower	Campaign	Required for late-game colony ship contact. Staged construction.

5.4 The Evolving World State

The colony meta is not a safe space. The planet pushes back. Several pressure systems run in the background, ensuring the meta layer carries genuine tension:

- Threat Escalation — hostile factions and dangerous fauna expand their territory over time. Neglected regions of the map become increasingly dangerous.
- Resource Depletion — heavily farmed resource nodes slowly diminish, requiring scouts to push into new territory to maintain factory throughput.
- Weather Events — seasonal cycles and random planetary events affect expedition conditions and occasionally the colony itself.
- Rival Factions — other human survivor groups operate independently, competing for resources and territory.

5.5 Meta Progression Currencies

Multiple currencies tick forward simultaneously, each with a different time horizon:

Currency	Source	Use
Raw Materials	Harvested in field	Factory input. Spent constantly.
Research Data	Discoveries and analysis	Unlocks new tech and buildings. Medium pacing.
Colony Reputation	Successful expeditions, trade	Unlocks faction relationships and late-game options.
Survey Progress	Map exploration	Unlocks planetary map features and final campaign content.
Scout Legacy Points	Veteran retirement, deaths	Unlocks memorial projects and cross-campaign bonuses.

6. The Planet

6.1 Biome Design

The planet surface is divided into distinct biome zones, each with its own visual identity, creature roster, resource profile, and environmental hazards. Biomes are distributed on a procedurally generated continental map that is consistent within a campaign.

Biome	Resources	Hazards	Creatures
Fungal Lowlands	Biological, spore compounds	Toxic bloom, disorientation	Swarm species, ambush predators
Crystal Mesa	Rare minerals, energy compounds	Unstable ground, sharp terrain	Territorial megafauna
Thermal Wastes	Geothermal materials, exotic metals	Heat damage, eruptions	Heat-adapted apex predators
Wetland Basin	Biological, water filtration	Reduced visibility, mire	Pack hunters, flying species
Ancient Ruins	Alien technology, research data	Structural collapse, hostile AI	Construct guardians
Deep Wilderness	All types (rare)	Extreme versions of all hazards	Unknown — must be discovered

6.2 Faction Ecosystem

The player is not alone on the planet. Multiple factions operate independently and react to the player's actions:

- Creature Factions — not a monolith. Different species have different behaviours, territories, and relationships with each other. Disrupting one species' territory may drive another into previously safe areas.
- Rival Survivors — other human colonial parties with their own agendas. Can be hostile, neutral, or allied depending on player choices.
- Ancient Constructs — remnant automated systems from whatever civilisation built the ruins. Neither hostile nor friendly by default — reactive to player actions in their territory.

6.3 The Planetary Threat

Each campaign features a central planetary threat — a macro-level antagonist that escalates throughout the game and provides the campaign's climactic challenge. The nature of the threat is partially procedural, giving each campaign a different identity.

The threat has a visible escalation track in the colony interface. Players can see it growing and understand that it will become a campaign-ending problem if ignored. This creates strategic pressure to push toward confronting it without making the game feel railroaded.

7. Art Direction & Presentation

7.1 Visual Style

Frontier Protocol uses an isometric perspective with a deliberately stylised art approach. The goal is strong readability and visual identity over photorealism. The art direction is achievable by a solo developer with a clear visual language.

The core visual contrast is: clean geometric human structures versus weird organic alien wilderness. Anything the colonists built is angular, functional, colour-coded. Everything the planet grew is asymmetric, bioluminescent, and visually unexpected.

Element	Direction
Colony / UI	Clean lines, functional palette, blueprint aesthetic. Confident and ordered.
Alien terrain	Organic, asymmetric, high colour contrast with the colony. Biomes have distinct palettes.
Scout characters	Strong silhouettes in isometric view. Personality readable from equipment and posture.
Portraits	Detailed character portraits used in dialogue, events, and the colony roster screen.
Lighting	Dynamic time-of-day and weather lighting. Bioluminescence at night. Storm visual effects.

7.2 Audio Direction

Audio design emphasises the alienness of the environment and the humanity of the scouts. The planet should sound alive and slightly wrong — familiar rhythms in unfamiliar timbres. The colony should sound like a functioning human space — mechanical, purposeful, warm.

Scout voice lines are a priority. Characters should comment on their situation, react to events, remark on their companions, and reflect their trait personalities. A scout with a dark humour trait sounds different from one with a fearful disposition, even if they're both reacting to the same creature attack.

7.3 Key Reference Points

- Hades (Supergiant Games) — isometric character expressiveness, reactive narrative, UI clarity.
- Darkest Dungeon — party morale system, character trauma, gothic atmosphere translated to a different tone.
- XCOM 2 — soldier personalisation, emotional investment in procedural characters, tactical encounter feel.
- Into the Breach — readability of tactical information, elegance of small-scale encounters.
- Wildermyth — character lifecycle narrative, emergent story from procedural traits.
- Divinity: Original Sin 2 — isometric free exploration combined with tactical combat transitions.

8. Monetisation Strategy

8.1 Philosophy

Frontier Protocol charges a fair upfront price for a complete game. There are no microtransactions, no loot boxes, no energy timers, and no pay-to-win elements of any kind. Every mechanical system is available to every player. Monetisation never creates an adversarial relationship with the player.

8.2 Pricing Model

Stage	Price	Content
Early Access	€7.99	Core loop complete. 2 biomes, 1 faction type, base scout system.
Version 1.0	€12.99	Full campaign, all biomes, complete scout lifecycle, full faction set.
Expansion (future)	€4.99	New biome, new planetary threat type, new scout specialisation tree.

8.3 Post-Launch Revenue

Post-launch content is sold as flat-price expansions — never as individual cosmetic items or randomised bundles. Players know exactly what they are getting before they purchase. Any cosmetic additions (scout appearance options, colony building skins) are bundled into expansions alongside mechanical content, never sold separately.

> **BUSINESS NOTE** *The goal is a player who feels the game was worth more than they paid for it. That player recommends it. Word of mouth is the primary marketing strategy.*

9. Technology

9.1 Engine — Godot 4

Frontier Protocol is built in Godot 4. The engine is free, open source, and has first-class C# support — making it accessible to a developer with a strong TypeScript and JavaScript background. Godot's scene and node architecture maps well to component-based thinking from web development. It exports natively to Windows, macOS, and Linux, covering Steam's primary platforms.

9.2 Technology Stack

Layer	Technology	Rationale
Engine	Godot 4	Free, open source, excellent 2D/isometric pipeline, C# support.
Language	C# (primary)	Strong typing, familiar to TypeScript devs, excellent Godot integration.
Rendering	Godot's built-in 2D renderer	TileMap system, isometric support, shader access.
Proc. generation	Custom C# systems	Chunk seeding, activity tables, event trees.
Save system	Godot serialisation + JSON	Campaign state, scout roster, colony state, map progress.
Distribution	Steam via Steamworks	Achievements, cloud saves, Workshop potential.

9.3 Key Technical Challenges

- Procedural chunk generation — chunks must be geographically stable across a campaign but vary in activity content. Requires a two-layer seeding system.
- Scout data persistence — the full scout lifecycle, trait history, relationship graph, and injury record must persist reliably across sessions and be efficiently queryable for event triggers.
- Isometric rendering order — sprite draw order in isometric view requires careful depth sorting. Well-understood problem with established Godot solutions.
- AI behaviour for creatures and factions — needs to feel dynamic and reactive without being computationally expensive. Behaviour trees with lightweight simulation in the background.

10. Development Roadmap

10.1 Guiding Principle

Build the core before building the content. The first milestone must produce something genuinely fun to play, even if it looks rough. Every subsequent milestone adds depth to a proven foundation. Scope is locked at each milestone — no new systems until the current ones are complete and stable.

> **SCOPE DISCIPLINE** *The game has failed if it is not fun in the first three months. Every hour spent on colony building, factory systems, or lore before the expedition loop is proven is a potentially wasted hour.*

10.2 Milestones

Phase	Duration	Deliverable	Success Criteria
0 — Foundation	1–2 months	Scout walks across isometric chunked map. Basic chunk transitions. Placeholder art throughout.	Movement feels good. Chunks load correctly.
1 — The Run	3–4 months	Combat encounters. Supply system. Basic scout stats and traits. Expedition debrief screen.	A full expedition is playable and tense.
2 — The Meta	3–4 months	Colony screen. Factory basics. Scout persistence across runs. Map persistence.	Running 5 expeditions in a row feels meaningful.
3 — Depth	4–6 months	Full scout lifecycle. Rescue missions. Faction behaviour. 2 biomes. Narrative events.	A 10-hour campaign is completable.
4 — Early Access	4–6 months	Polish, balance, art pass, audio. Steam page live. Early access launch.	Positive reviews. Active player feedback loop.
5 — Version 1.0	6–12 months	Full content. Campaign complete. All systems finalised. 1.0 launch.	Full critical reception. Expansion planning begins.

10.3 Immediate Next Steps

Before any game development begins, the following should be in place:

- Install Godot 4 and complete a basic isometric movement tutorial project.
- Play Wildermyth, Invisible Inc, and Darkest Dungeon 2 with a design notebook open.
- Establish a design journal — a living document where all ideas, changes, and decisions are recorded.
- Define the first prototype goal precisely: a scout moving across three connected chunks with basic interaction.

- Set a hard three-month checkpoint: if the movement and encounter feel is not fun, reassess before building anything else.

Frontier Protocol — GDD v0.1

Living document. Update with every significant design decision.